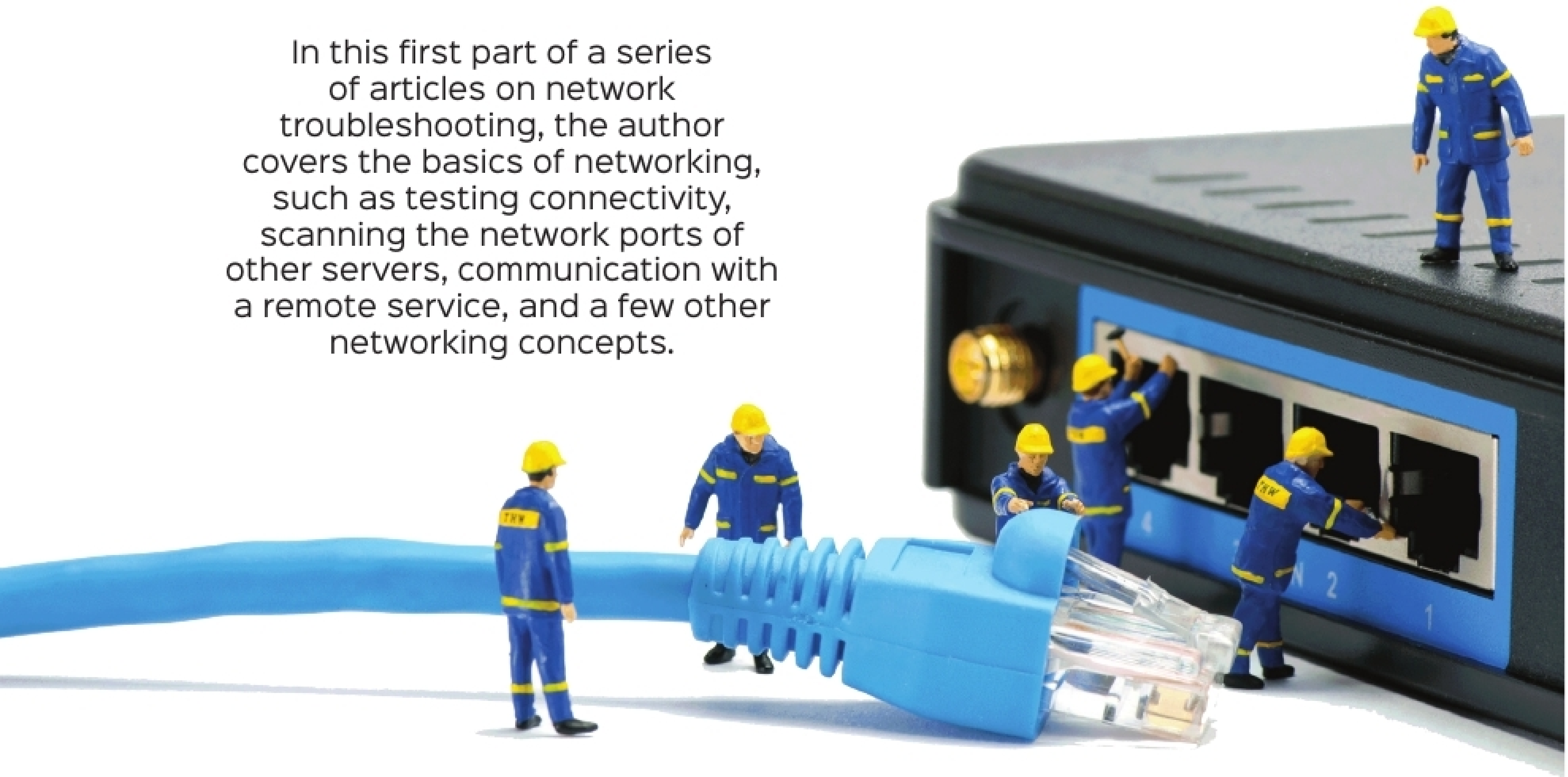
Troubleshooting Network Issues: An Introduction

In this first part of a series of articles on network troubleshooting, the author covers the basics of networking, such as testing connectivity, scanning the network ports of other servers, communication with a remote service, and a few other networking concepts.



The Internet Control Message Protocol (ICMP) is a low-level protocol used to test host availability and send error messages. One of the first steps in testing for connectivity is to send ICMP echo requests to the remote host. Hosts, by default, are configured to send an ICMP echo reply to indicate that they are present and running. This is accomplished with the *ping* command.

The *ping* command takes the host name, or IP address, of the host of interest as an argument. When the *-b* option is used, the command argument specified is a broadcast address. By default, the ping will continuously send ICMP echo requests every second. All responses that are received are displayed with their packet sequence number and latency time. When the user interrupts the command with a *Ctrl+C*, then the *ping* command displays a summary. You can refer to the output in Figure 1 for the *ping* command output.

```
[root@ipa ~]# ping google.com
PING google.com (216.58.203.110) 56(84) bytes of data.
64 bytes from syd09s15-in-f14.1e100.net (216.58.203.110): icmp_seq=1 ttl=50 time=315 ms
64 bytes from syd09s15-in-f14.1e100.net (216.58.203.110): icmp_seq=2 ttl=50 time=275 ms
64 bytes from syd09s15-in-f14.1e100.net (216.58.203.110): icmp_seq=3 ttl=50 time=219 ms
64 bytes from syd09s15-in-f14.1e100.net (216.58.203.110): icmp_seq=4 ttl=50 time=229 ms
64 bytes from syd09s15-in-f14.1e100.net (216.58.203.110): icmp_seq=5 ttl=50 time=265 ms
64 bytes from syd09s15-in-f14.1e100.net (216.58.203.110): icmp_seq=6 ttl=50 time=352 ms
64 bytes from syd09s15-in-f14.1e100.net (216.58.203.110): icmp_seq=7 ttl=50 time=388 ms
^C
--- google.com ping statistics ---
8 packets transmitted, 7 received, 12% packet loss, time 7008ms
rtt min/avg/max/mdev = 219.994/292.380/388.255/57.902 ms
[root@ipa ~]# █
```

Figure 1: Ping output

There are a couple of options that make the ping command very useful in a shell program. The *-c* counts the number of echo requests to send out. The *-W* timeout specifies how many seconds it has to wait before timing out. The ping command in Figure 2 is sending two echo requests and is waiting five seconds for a reply.

```
[root@ipa ~]#  ping -c 2 -W 5 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=86.1 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=87.5 ms

--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 86.166/86.854/87.542/0.688 ms
[root@ipa ~]# █
```

Figure 2: Ping command sending two echo requests and waiting five seconds for a reply

Ping returns a zero exit status when the target host responds and returns a non-zero exit status when the target is not reachable.

The *ping6* command is used to send echo requests to an IPv6 address. For IPv6, when doing a ping to a link-local scope address, a link specification using the *-I* option is required. View the following command for the same:

```
# ping6 -I eth0 <IPv6-address>
```